Natdanai (Bank) Leelathanapipat

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Education

University of Waterloo — Waterloo, ON

Bachelor of Applied Science, Computer Engineering (Co-op) — *Sept. 2025 – Apr. 2030*

- Relevant coursework: Algorithms and Data Structures (C++), Digital Computers (architecture, assembly, memory systems), Digital Circuits and Systems (Verilog), Fundamentals of Programming (C++).

Work Experience

Software Developer (Co-op) — May 2026 – Aug. 2026

Digital Health Center, Faculty of Medicine, Chulalongkorn University — *Bangkok, Thailand*

- Built and shipped a driver dispatch platform whose assignment engine is a deterministic constraint solver over rotation ledgers, rest gaps, and distance-based co-driver rules (Next.js 16, TypeScript, Prisma, PostgreSQL); sole developer, now in daily production use dispatching ∼18 trips/week across the faculty's six-driver fleet, replacing a manual scheduling process.
- Deployed it to hospital Ubuntu servers end to end — nginx reverse proxy, systemd, TLS, UFW — and validated the solver across 100,000+ simulated scheduling days plus an adversarial pre-deploy audit.
- Built a camera-based CPR coach on pose estimation (YOLO-pose, MediaPipe, PyQt6) grading compression depth, rate, and posture against AHA 2020 in real time; torso-normalized ratios removed pixel-to-cm calibration, reaching 98.7% agreement with measured ground truth in testing with 82 trainees.

Research Assistant (part-time) — Jul. 2024 – May 2025

King Mongkut's Institute of Technology Ladkrabang — *Bangkok, Thailand*

- Built a real-time reservoir monitoring system for proactive flood management, instrumenting dam substations with OTT C31 current meters and float level switches to capture water level on both sides of the dam door and flow velocity through it.
- Designed the substation-to-main-station telemetry link — 433 MHz Yagi-Uda antenna with HT12E/HT12D encoding for secure transmission — with an Arduino Mega 2560 processing readings and a Raspberry Pi serving them to a live web dashboard; validated in both controlled and field conditions.
- Co-authored and presented the resulting paper at the 2024 IEEE Asian Conference on Communication and Networks in Bangkok; indexed in IEEE Xplore.

Software Engineering Intern — Jul. 2024 – Aug. 2024

ODDS-Thailand — *Bangkok, Thailand*

- Optimized MongoDB queries across a 20M+ record cluster with compound indexes and refactored aggregation pipelines, cutting dashboard retrieval time by 45%.
- Developed responsive React and Tailwind CSS components for a production financial services platform serving 200K+ users, working inside an existing codebase under code review.

Projects

Pood (Offline Thai/English Voice Dictation) — 2026 – Present

Python, Whisper, Silero VAD, Windows UI Automation

- Built a free, fully local dictation and voice-control app for Windows and macOS that routes each utterance to the best Whisper model for its language and drives the computer by voice, down to clicking on-screen elements by number.
- Added timing-aware "enter" detection from VAD silence, a hallucination guard, and a benchmark and LoRA fine-tuning pipeline that scores Thai, English, and code-switched speech separately.

Badminton Match Analysis (Computer Vision & ML) — 2026 – Present

github.com/Bank-Leela

- Adapted a TrackNetV3 fork for shuttlecock tracking on BWF broadcast footage, deliberately scoping to matches outside the model's original training data to measure generalization rather than report in-distribution accuracy.
- Trained a LightGBM classifier on a hand-labeled dataset to grade shot quality as good, risky, or bad, pairing court-position and rally-context features to surface shot-selection patterns coaches can act on.

Sentinel (AI Fraud Detection Platform) — Mar. 2026

GenAI Genesis, Canada's Largest AI Hackathon — *Toronto, Canada*

- Built a real-time fraud detection platform (FastAPI, Next.js, TypeScript) end to end, with live transaction triage, explainable risk scoring, and graph-based network analysis so analysts could act on alerts, not just read scores.

Technical Skills

Languages: C++, C, Python, JavaScript, TypeScript, Dart, SQL, Verilog, HTML/CSS
Frameworks: React, Next.js, Node.js, FastAPI, Prisma, Tailwind CSS, PyQt6, Flutter
Data & ML: PostgreSQL, MongoDB, MediaPipe, YOLO, OpenCV, Whisper, LightGBM, NumPy, Pandas
Infrastructure & Tools: Linux, nginx, systemd, Docker, Git, Vitest, Pytest, Vercel, REST APIs, Arduino, Raspberry Pi
Activities: UW Badminton Club; speedcubing; founder of PuzzlePeace (cube workshops for 50+ hospitalized children)